جامعة التقنية والعلوم التطبيقية University of Technology and Applied Sciences Muscat مسقط	**UNIVERSITY OF TECHNOLOGY AND APPLIED SCIENCES**
	Mathematics Section
	Quiz II **Semester: 1, A. Y.: 2024 / 25**
	Date: 27,11,2024 **Marks: 7.5** **Version A**
Student Name:	[redacted]
Student ID:	[redacted]

Course Name	Calculus I
Course Code	MATH1200/ MATH 1202
Section	11

1.5

Excellent

Question: 1 **[1.5 Marks]**

Find the derivative for the function $f(x) = \overset{out}{\sin}(9 \overset{in}{-} 2x^5) + \overset{out}{\ln}(\overset{in}{7x})$

$f'(x) = \cos(9-2x^5) \cdot -10x^4 + \frac{1}{7x} \cdot 7$ #

1.5

Question: 2 **[2 Marks]**

The radius of a sphere is increasing at a rate of 3 m/sec. How fast is the volume changing when the radius is 6 m?

$\frac{dr}{dt} = 3$

$\frac{dv}{dt} = ?$

$r = 6$

Volume $= \frac{4}{3}\pi r^3$

$\frac{dv}{dt} = 4\pi r^2 \cdot \frac{dr}{dt}$

$\frac{dv}{dt} = 4\pi(6)^2 \cdot 3$

$\frac{dv}{dt} = 432\pi$ #

2

$\frac{4}{3}\pi r^2$

Question: 3 **[2 Marks]**

Find $\frac{dy}{dx}$ if $3y + y^2 = x^2 y$

$3y' + 2yy' = 2x \cdot y + y' \cdot x^2$

$3y' + 2yy' - y'x^2 = 2x \cdot y$

$y'[3 + 2y - x^2] = 2x \cdot y$

$y' = \frac{2x \cdot y}{3 + 2y - x^2}$

2

Question: 4 **[2 Marks]**

Find the absolute extreme values of the function $f(x) = \frac{x^2}{x^2+9}$ on the interval $-1 \le x \le 2$

$x^2 \cdot (x^2+9)^{-1}$

$f'(x) = \frac{2x(x^2+9) - (2x \cdot x^2)}{(x^2+9)^2} = 0$

~~$f'(x) = 2x \cdot (x^2+9)^{-1} - (x^2+9)^{-2} \cdot 2x = 0$~~

~~$2x \cdot (x^2+9)^{-1} - (x^2-9)^{-2} \cdot 2x = 0$~~

step1

$\frac{2x^3 + 18x - 2x^3}{(x^2+9)^2} = 0$

$\frac{18x}{18} = \frac{0}{18}$

$x = 0$ ✓ critical point

step2 $\sqrt{(x^2+9)^2} = \sqrt{0}$

$x^2 + 9 = 0$

$x^2 = -9$

$x =$ error

2

-1	0	2
0.1	0	0.3

$f(-1) = \frac{(-1)^2}{(-1)^2+9}$

abs max = $(2, 0.3)$

abs min = $(-1, 0.1)$

$(0, 0)$

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	Mathematics Section
	Quiz III **Semester: 1, A. Y.: 2024 / 25**
	Date: 11.12.2024 **Marks: 7.5** **Version B**
Student Name:	[redacted]
Student ID:	[redacted]

Course Name	Calculus I
Course Code	MATH1200/ MATH 1202
Section	11

6.5

Question: 1 **[1.5 Marks]**

If the function $f(x) = kx^2 + 2x - 6$ satisfies the Rolle's theorem over the interval $[1, 7]$. Determine the value of k.

use 3rd condition

$f(a) = f(b)$

$f(1) = f(7)$

$k(1)^2 + 2(1) - 6 = k(7)^2 + 2(7) - 6$

$k - 4 = 49k + 8$

$k - 49k = 8 + 4$

$\frac{-48k}{-48} = \frac{12}{-48}$

$k = -\frac{1}{4}$

1.5

Question: 2 **[2 Marks]**

Verify Mean value theorem for the function $f(x) = (x-3)^2$ over the given interval $[1,4]$.

- Continous
- exist
- $f'(c)$

$f(x)$ continous on $[1,4]$

$f'(x) = 2(x-3) \cdot 1$ exist on $[1,4]$

$f'(c) = \frac{f(b) - f(a)}{b - a}$

$2(c-3) = \frac{f(4) - f(1)}{4 - 1} = \frac{1-4}{3} = -1$

$f(4) = (4-3)^2 = 1$

$f(1) = (1-3)^2 = 4$

$\frac{2(c-3)}{2} = \frac{-1}{2}$

$c - 3 = -\frac{1}{2}$

$c = \frac{5}{2} \approx 2.5$ it's on $(1,4)$

$\therefore$ verived ✓

2

Question: 3 **[2 Marks]**

Find two positive integers such that their sum is 12 and minimizes the sum of their squares.

$x + y = 12 \longrightarrow y = 12 - x$

min $F = x^2 + y^2$

$F' = 2x + 2(12-x) \cdot -1$

$= 2x + (24 - 2x) \cdot -1$

$-2x + -22x = 0$

$x(2 - 22) = 0$

Question: 4 **[2 Marks]**

Use L'Hospital's rule to find the limit of $\lim_{x\to 0} \frac{\cos x - 1}{e^x - x - 1}$ $= \frac{\cos 0 - 1}{e^0 - 0 - 1} = \frac{0}{0}$

$\lim_{x\to 0} \frac{-\sin x}{e^x - 1} = \frac{-\sin 0}{e^0 - 1} = \frac{0}{0}$

$\lim_{x\to 0} \frac{-\cos x}{e^x} = \frac{-1}{1} = -1$

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	Mathematics Section
	Quiz I **Semester: 1, A. Y.: 2024 / 25**
	Date: / **Marks: 7.5** / **Version B**
Student Name	[redacted]
Student ID	[redacted]

Course Name	Calculus I
Course Code	MATH1200/ MATH 1202
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7.5 Excellent!

Question: 1 **[1+0.5+0.5 Marks]**

(a) Does $\lim_{x\to 2} f(x)$ exists? Give reason.

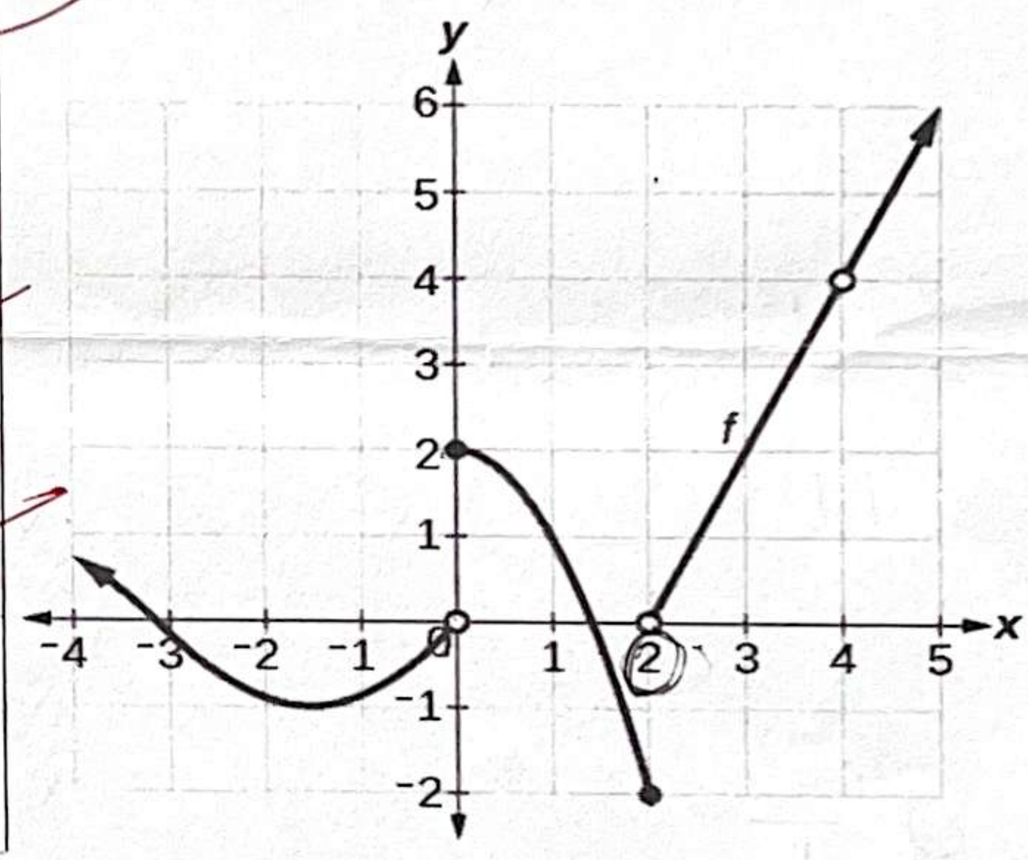
LHL $= \lim_{x\to 2^-} = -2$

RHL $= \lim_{x\to 2^+} = 0$

RHL $\neq$ LHL Does Not Exist

(b) $\lim_{x\to 0^+} f(x) =$ 2

(c) $f(2) =$ -2



Question: 2 **[2 Marks]**

Evaluate: $\lim_{x\to 4} \dfrac{x-4}{\sqrt{x+12}-4}$ $\quad \dfrac{4-4}{\sqrt{4+12}-4} = \dfrac{0}{0}$ type 2

$\lim_{x\to 4} \dfrac{x-4}{\sqrt{x+12}-4} \times \dfrac{\sqrt{x+12}+4}{\sqrt{x+12}+4}$

$\lim_{x\to 4} \dfrac{(x-4)(\sqrt{x+12}+4)}{x+12-16}$

$\lim_{x\to 4} \dfrac{\cancel{x-4}(\sqrt{x+12}+4)}{\cancel{x-4}}$

$\lim_{x\to 4} \sqrt{4+12}+4 = 8$ #

2

Question: 3 **[2 Marks]**

If the function $f(x) = \begin{cases} 1-2a & x<2 \\ 11 & x=2 \\ 2b-1 & x>2 \end{cases}$ is continuous everywhere, then find the value of 'a' and 'b'

$-$ $+$

$< 2 >$

$f(x) = \lim_{x\to 2^-} = \lim_{x\to 2^+}$

$11 = \boxed{1-2a} = \boxed{2b-1}$

2

$1-2a = 11$ $\quad$ $2b-1 = 11$

$\frac{-2a}{-2} = \frac{10}{-2}$ $\quad$ $\frac{2b}{2} = \frac{12}{2}$

$a = -5$ # $\quad$ $b = 6$ #

Question: 4 **[1.5 Marks]**

Find the horizontal and vertical asymptotes of the function $f(x) = \dfrac{3x^2+3x-36}{x^2-9}$

1-1. A

$m = n \rightarrow y = \frac{a}{b}$

1.5

$y = \frac{3}{1} = 3$

$\lim_{x\to\infty} \dfrac{\frac{3x^2}{x^2} + \frac{3x}{x^2} - \frac{36}{x^2}}{\frac{x^2}{x^2} - \frac{9}{x^2}}$

$\lim_{x\to\infty} \dfrac{3 + \frac{3}{x} - \frac{36}{x^2}}{1 - \frac{9}{x^2}}$

$\frac{3+0-0}{1-0} = \frac{3}{1} = 3$

V. A

$x^2 - 9 = 0$

$x^2 = 9$

$x = \pm 3$

check:

$x=+3$ $\quad 3(3)^2 + 3(3) - 36 = 0$ x

$x=-3$ $\quad 3(-3)^2 + 3(-3) - 36 = -18$ ✓

$x = -3$